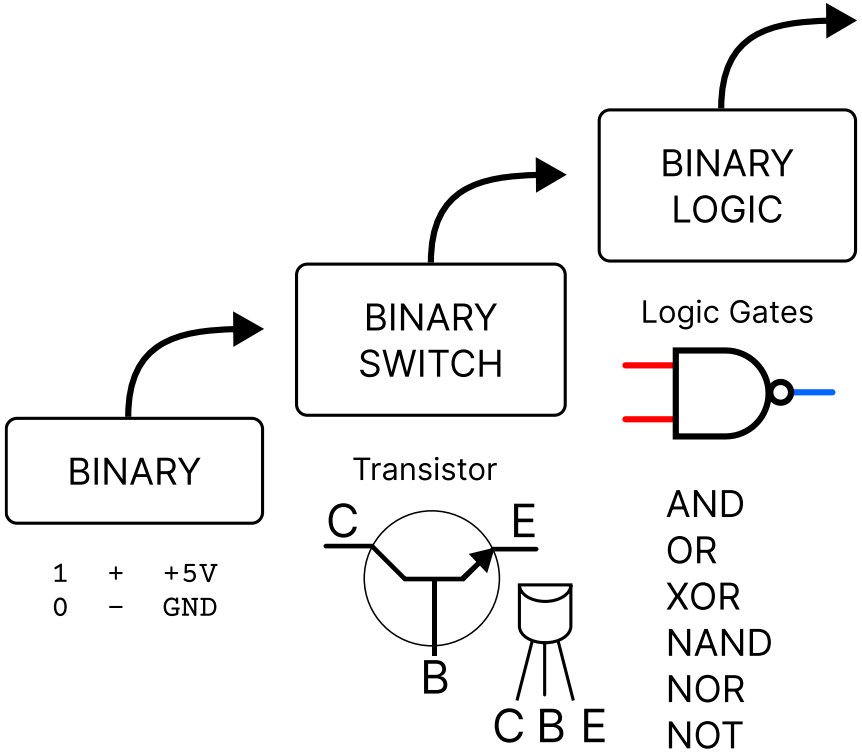
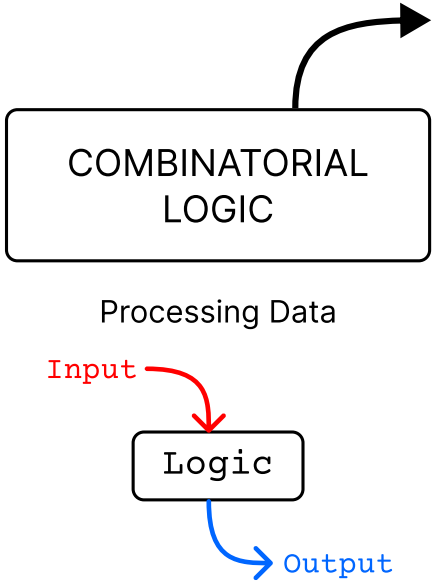
COSC204
Lectures 1-5

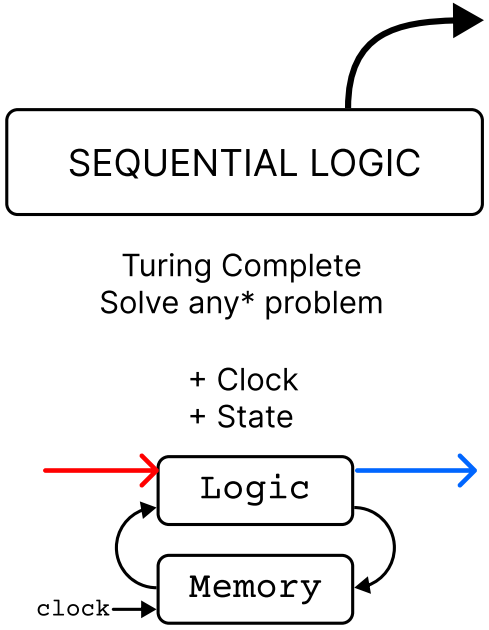




Input/output can be **ANY** data format.

Processing logic can be **ANY** function.

- add numbers
- check parity
- autotune audio
- image filtering

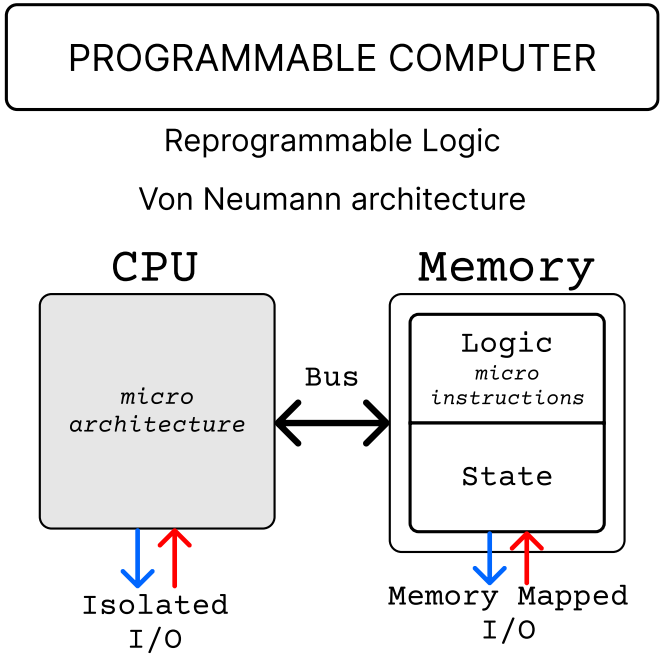


Memory stores state
- Latches
- Flip-flops

Simulated "State Machine" is "Turing Complete"

It can solve any* problem, including recursive problems, combinatorial logic can't

**cannot solve undecidable problems e.g. "Halting Problem". Mathematical paradoxes.*

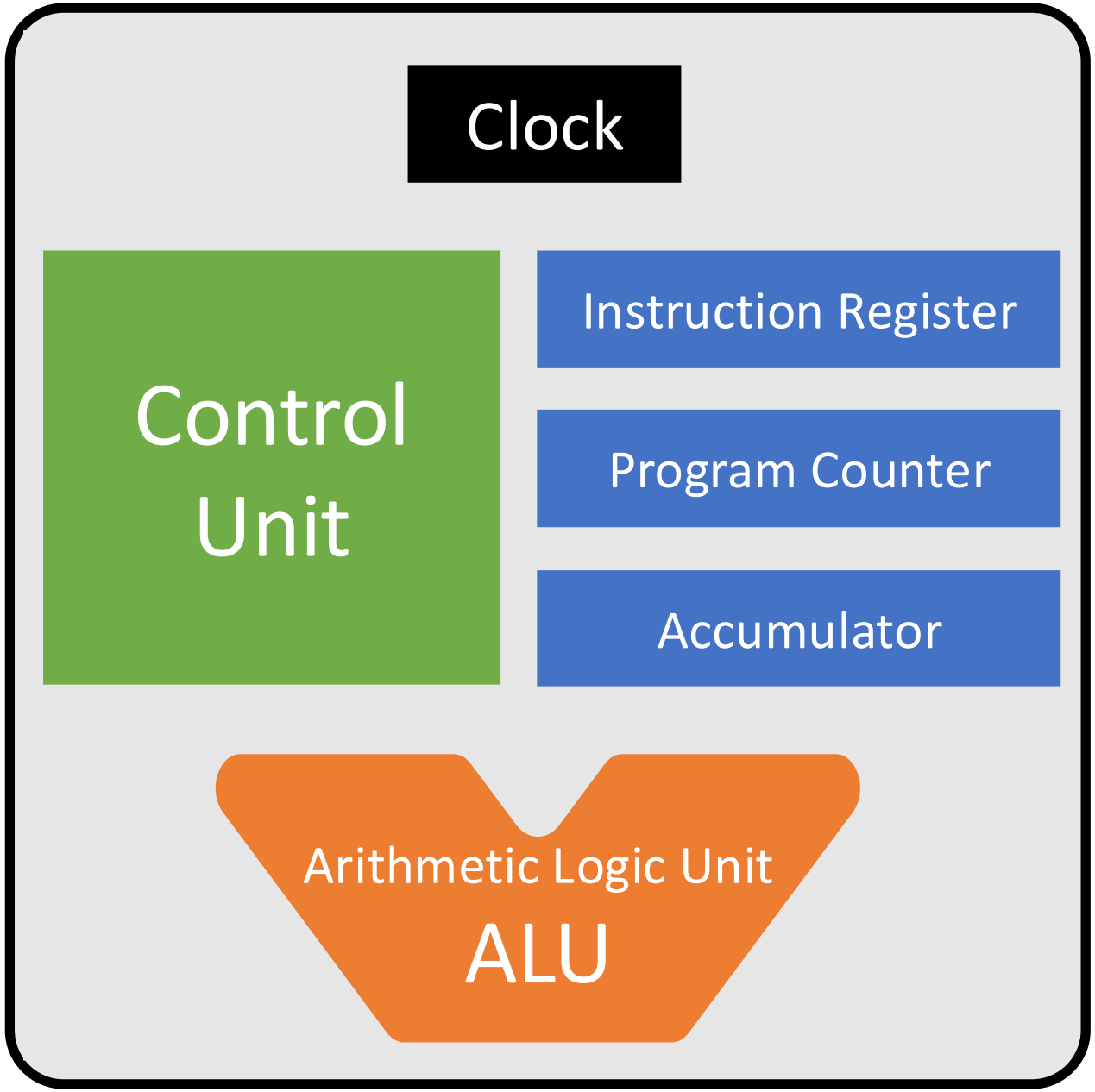


Store logic in memory.
Logic can be changed for solving different problems.

Program can't be logic gates, but instead *micro instructions*.

CPU can execute the instructions in memory.

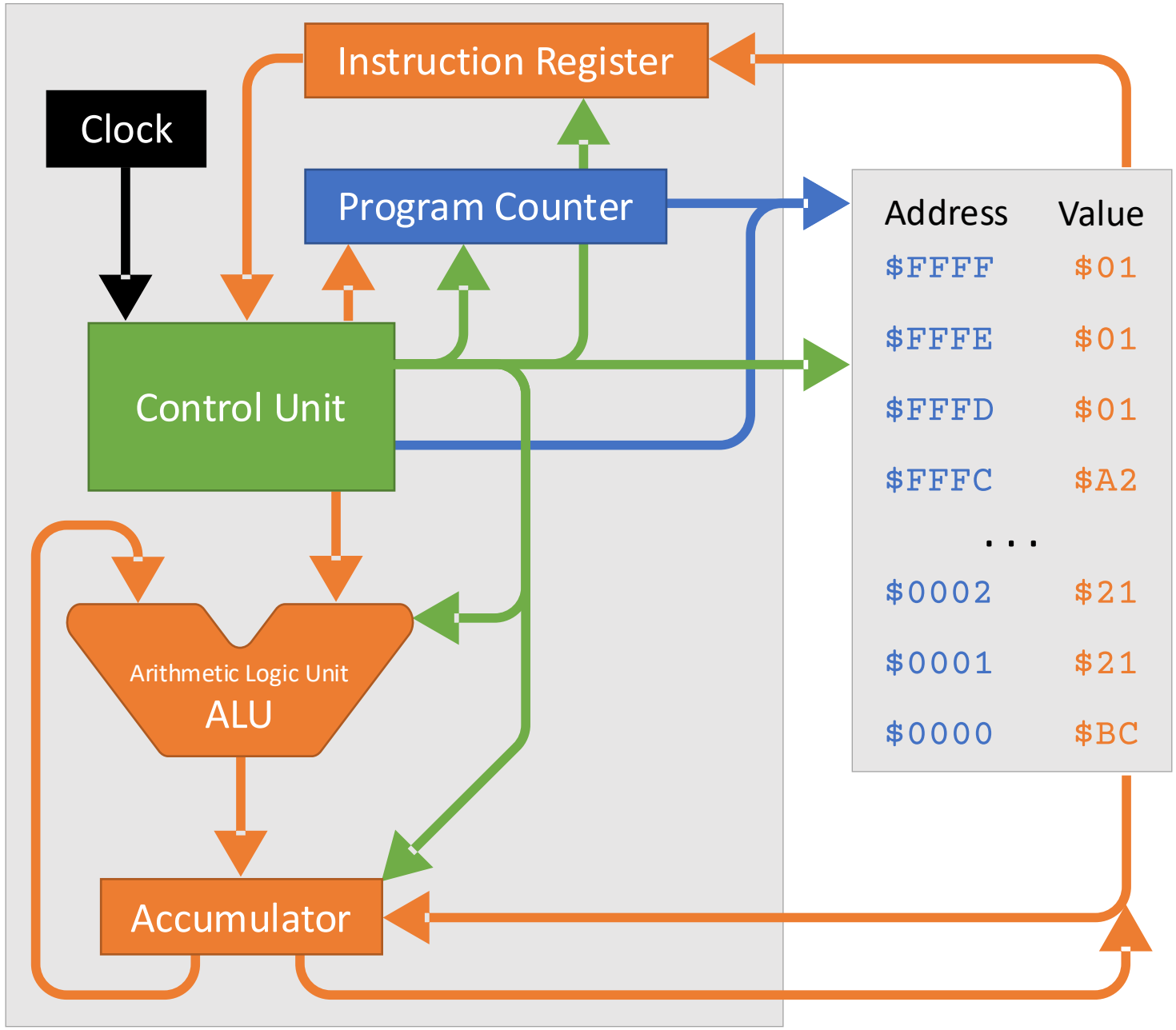
- Machine code `0100 1010`
- Assembly `LDA #$A0`
- C `a = 10;`


Clock
Control Unit
Instruction Register
Program Counter
Accumulator
Arithmetic Logic Unit
ALU

Control Bus

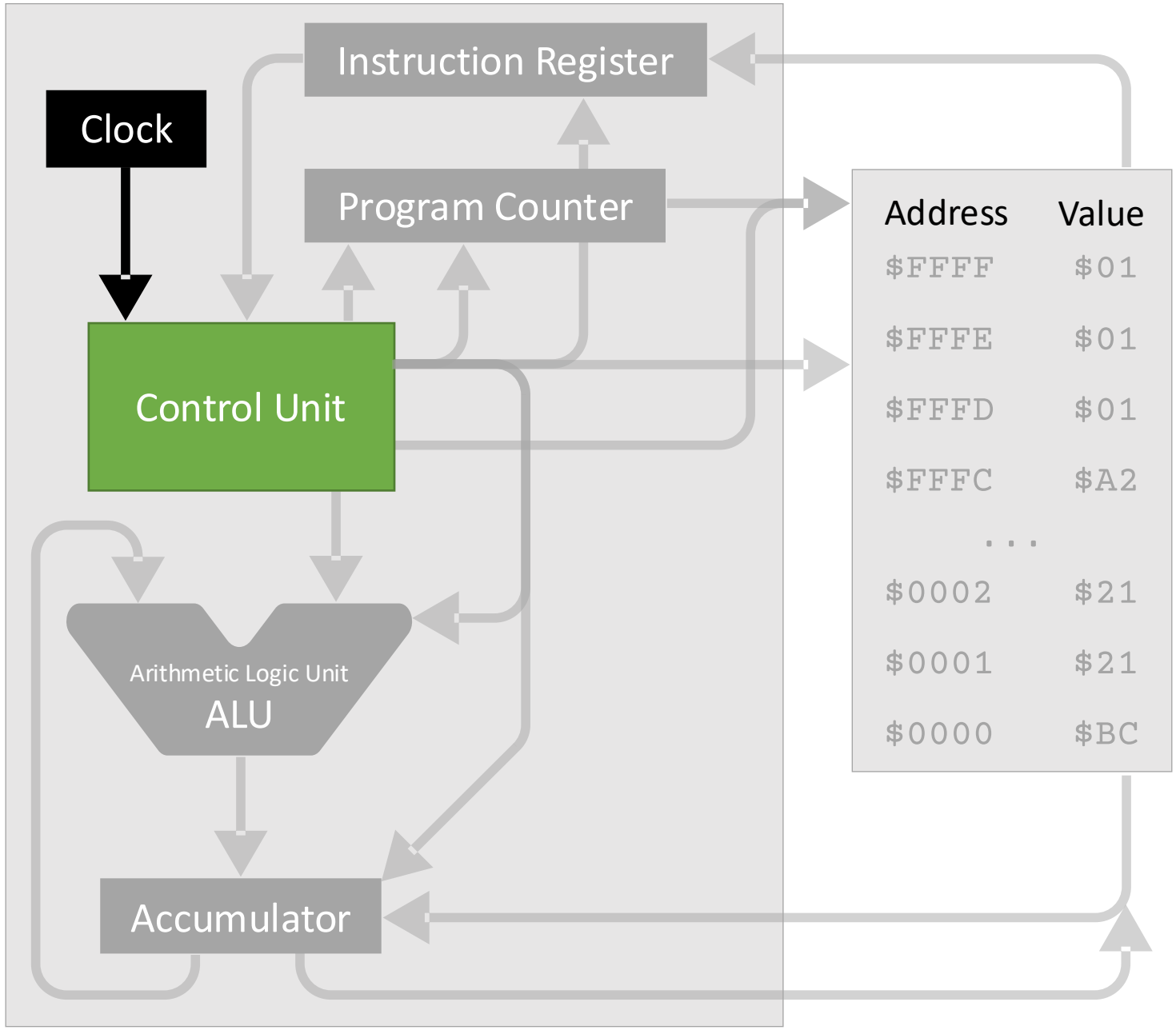
Address Bus

Data Bus



Fetch Instruction

Clock pulse



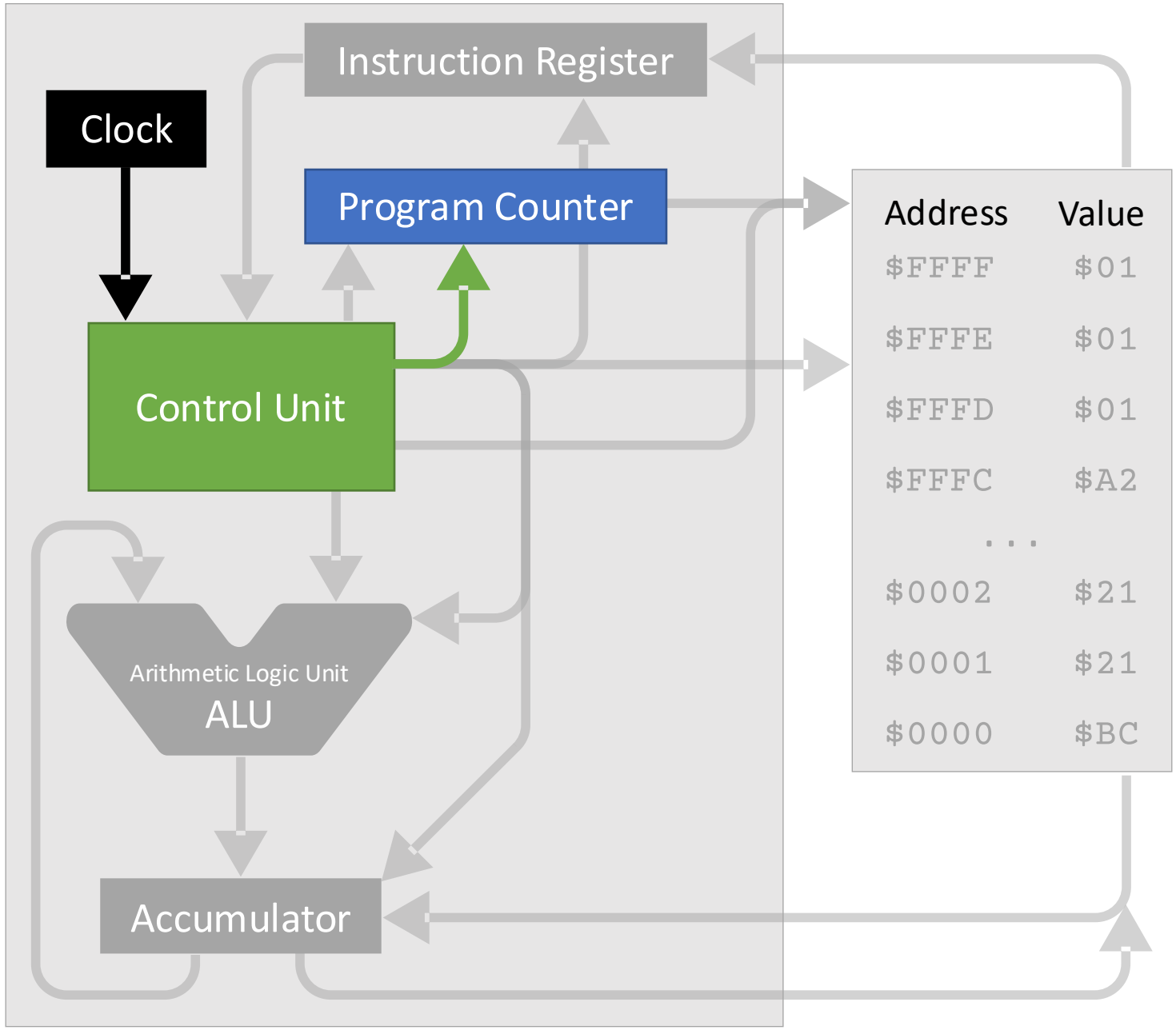
Control Bus

Address Bus

Data Bus

Fetch Instruction

Clock pulse
CU enables PC



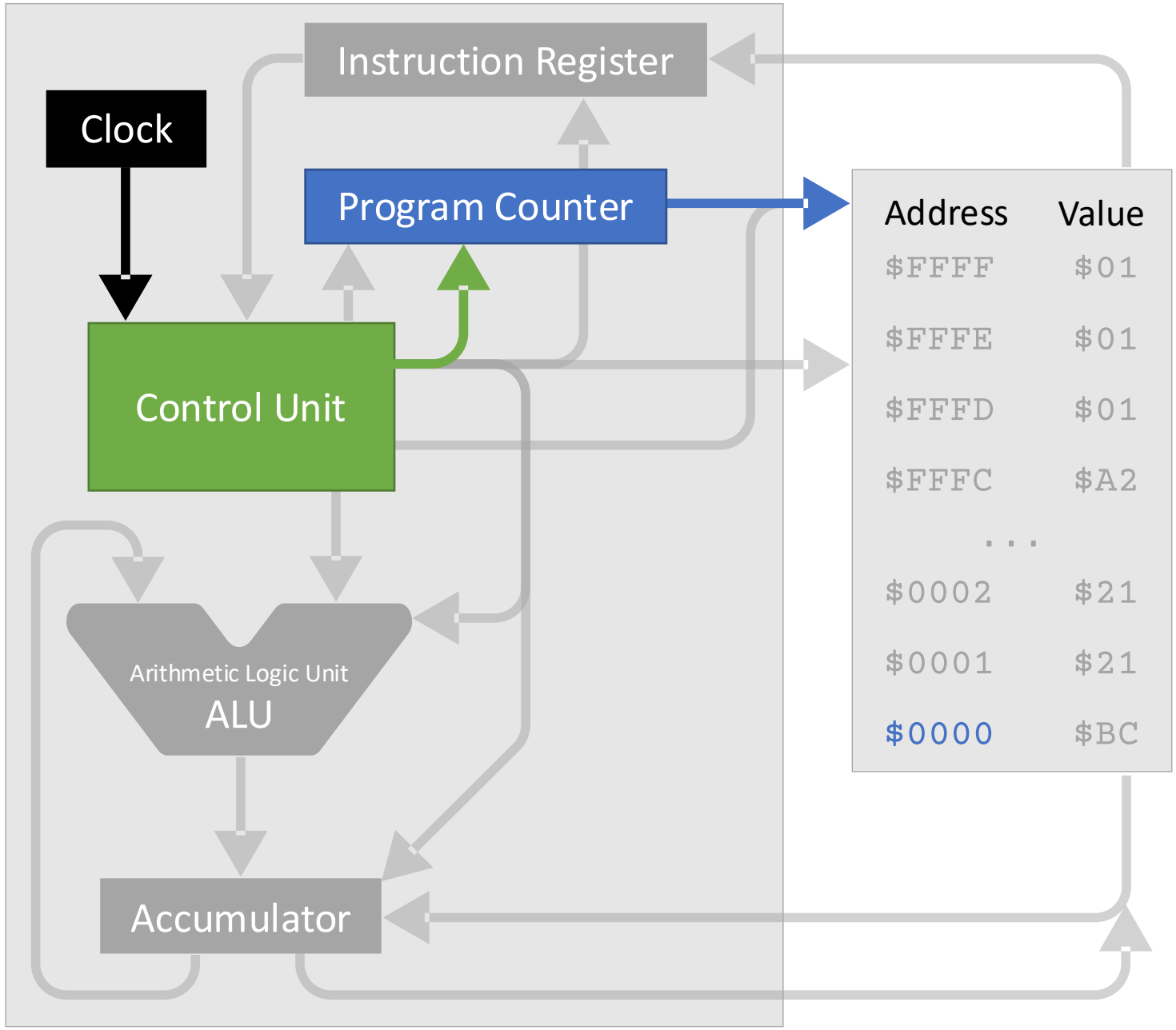
Control Bus

Address Bus

Data Bus

Fetch Instruction

Clock pulse
CU enables PC
PC selects address



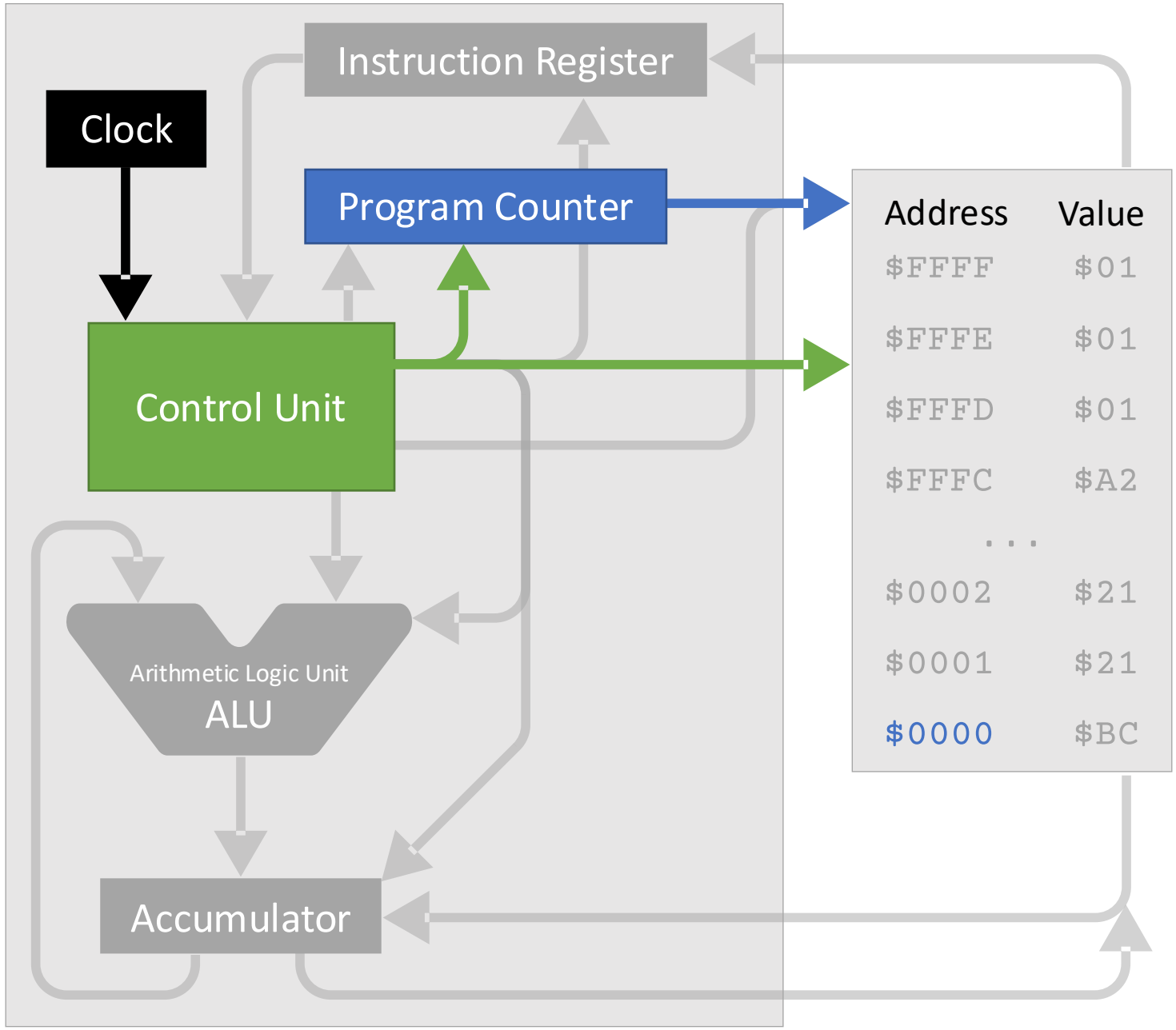
Control Bus

Address Bus

Data Bus

Fetch Instruction

Clock pulse
CU enables PC
PC selects address
CU enables memory read



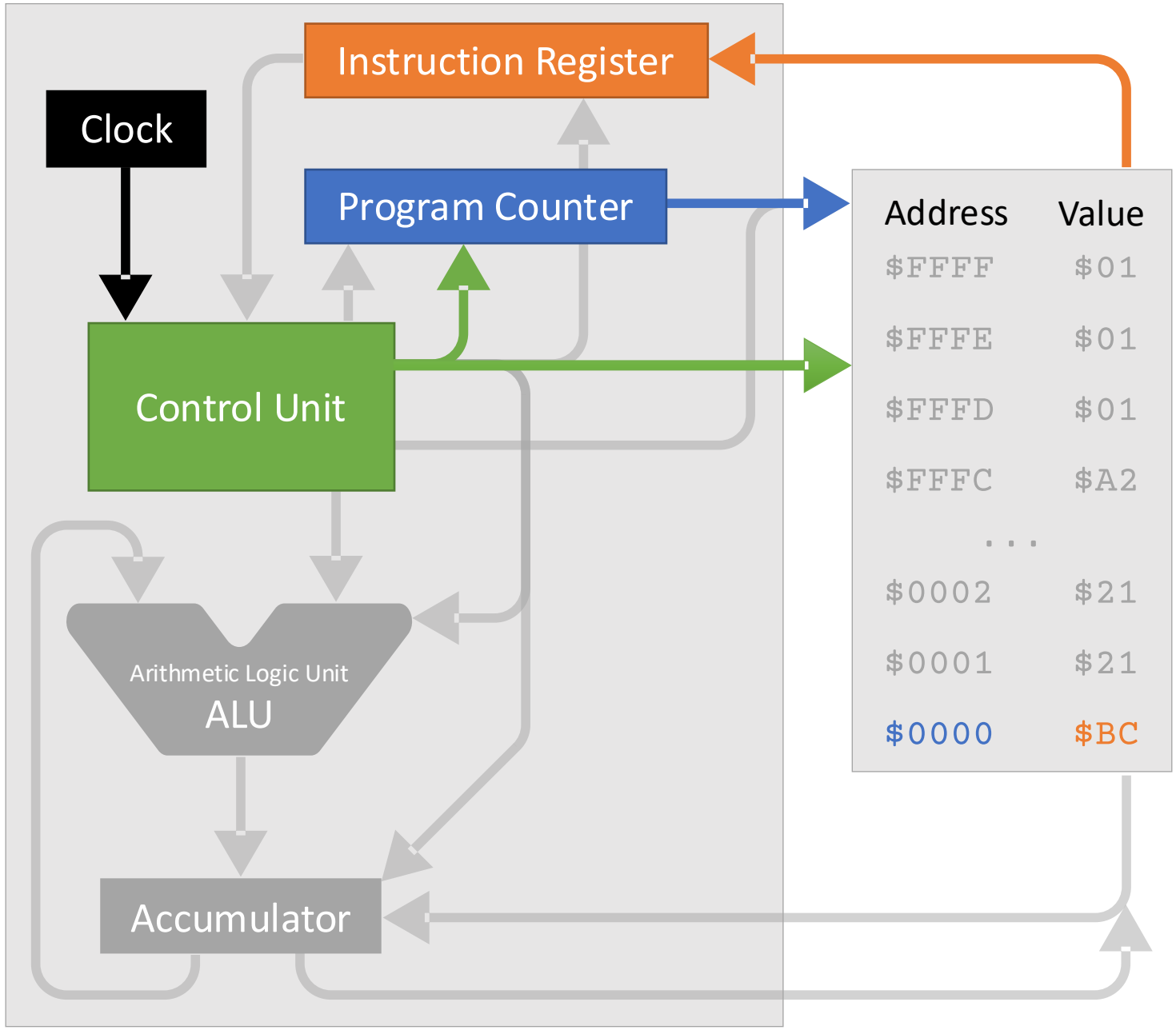
Control Bus

Address Bus

Data Bus

Fetch Instruction

Clock pulse
CU enables PC
PC selects address
CU enables memory read
IR reads value



Control Bus

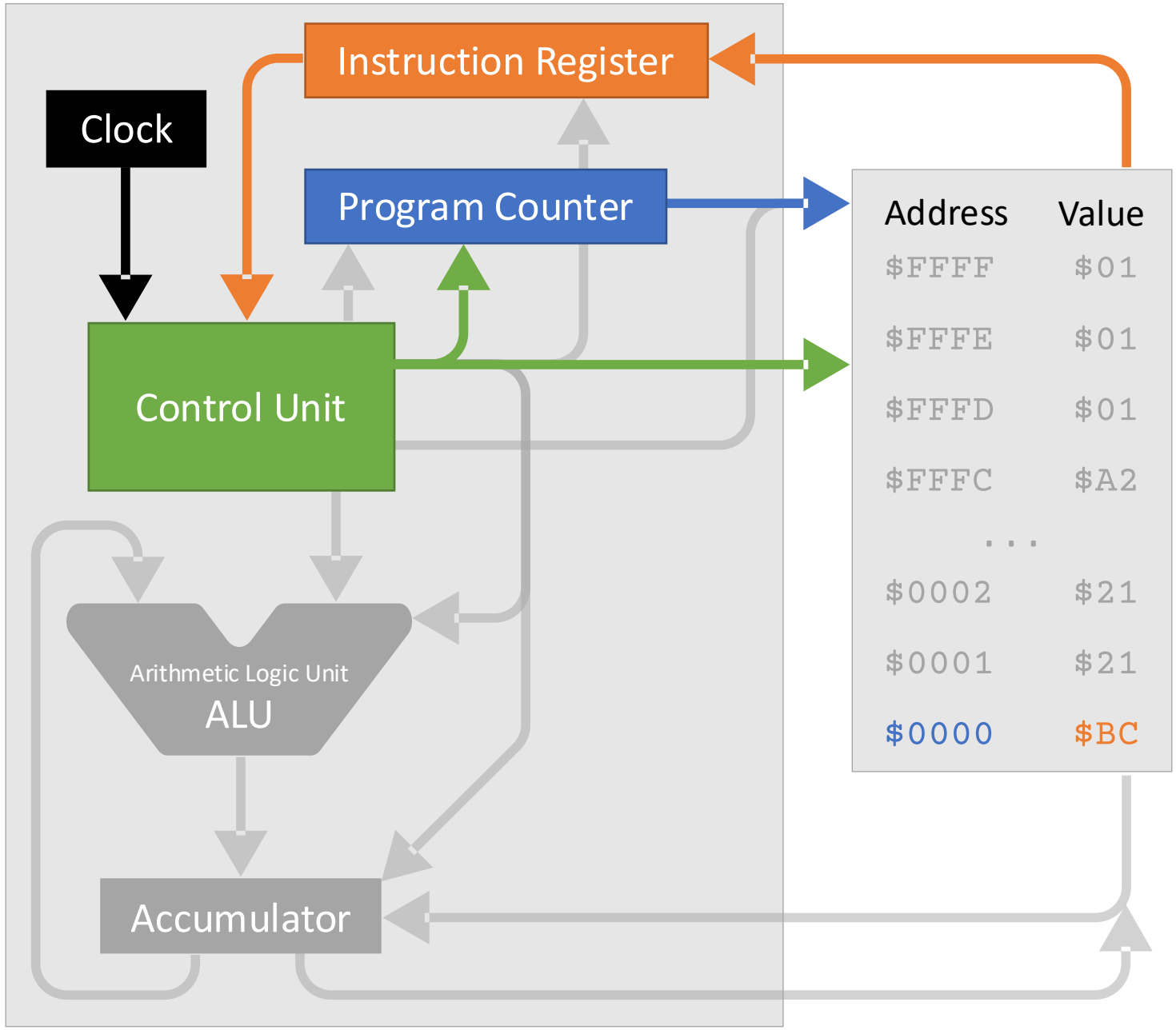
Address Bus

Data Bus

Fetch Instruction

Clock pulse
CU enables PC
PC selects address
CU enables memory read
IR reads value
CU receives IR

Occurs simultaneously!



Control Bus

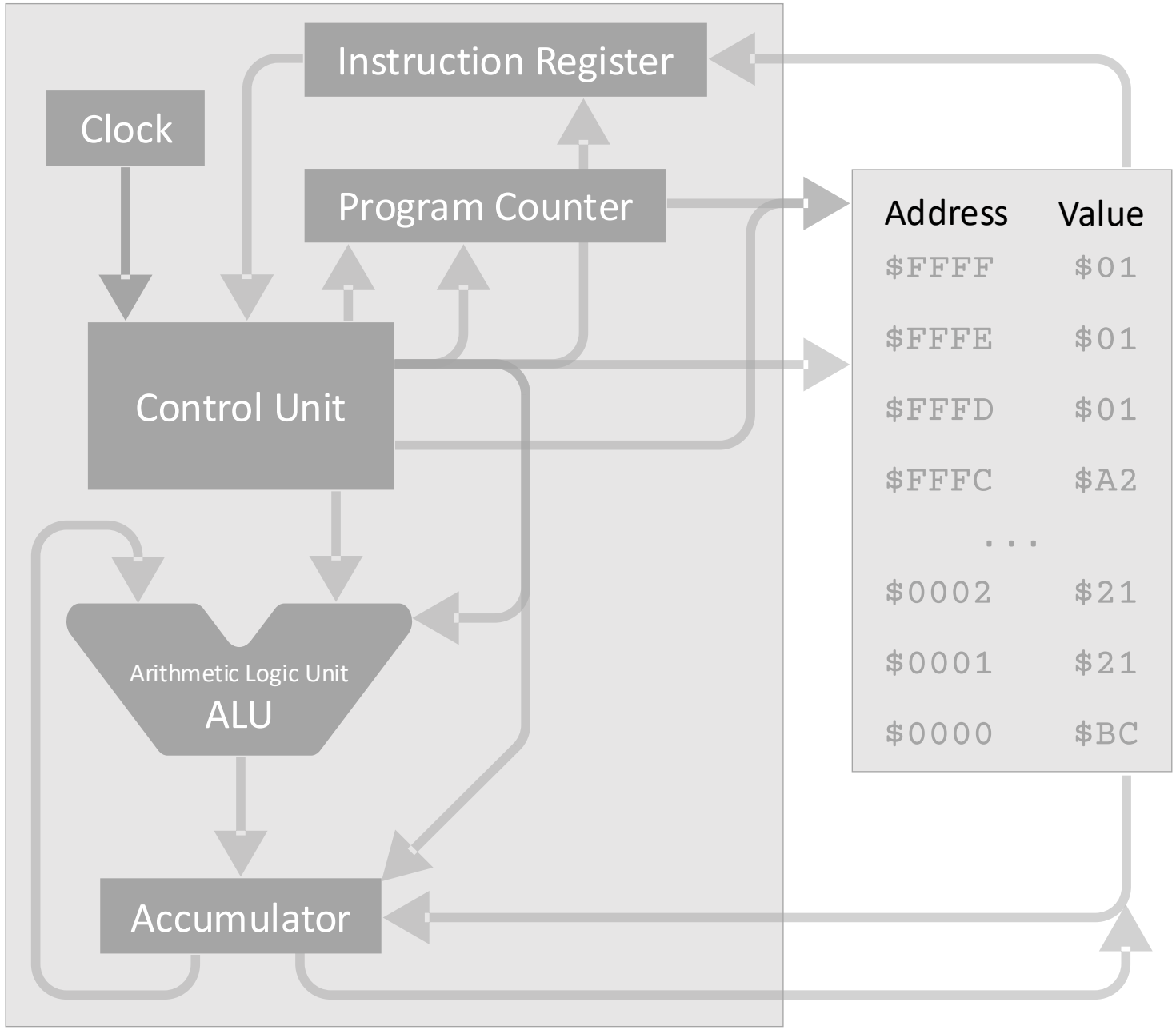
Address Bus

Data Bus

Fetch Instruction

Clock off

Next clock pulse will trigger CU to execute



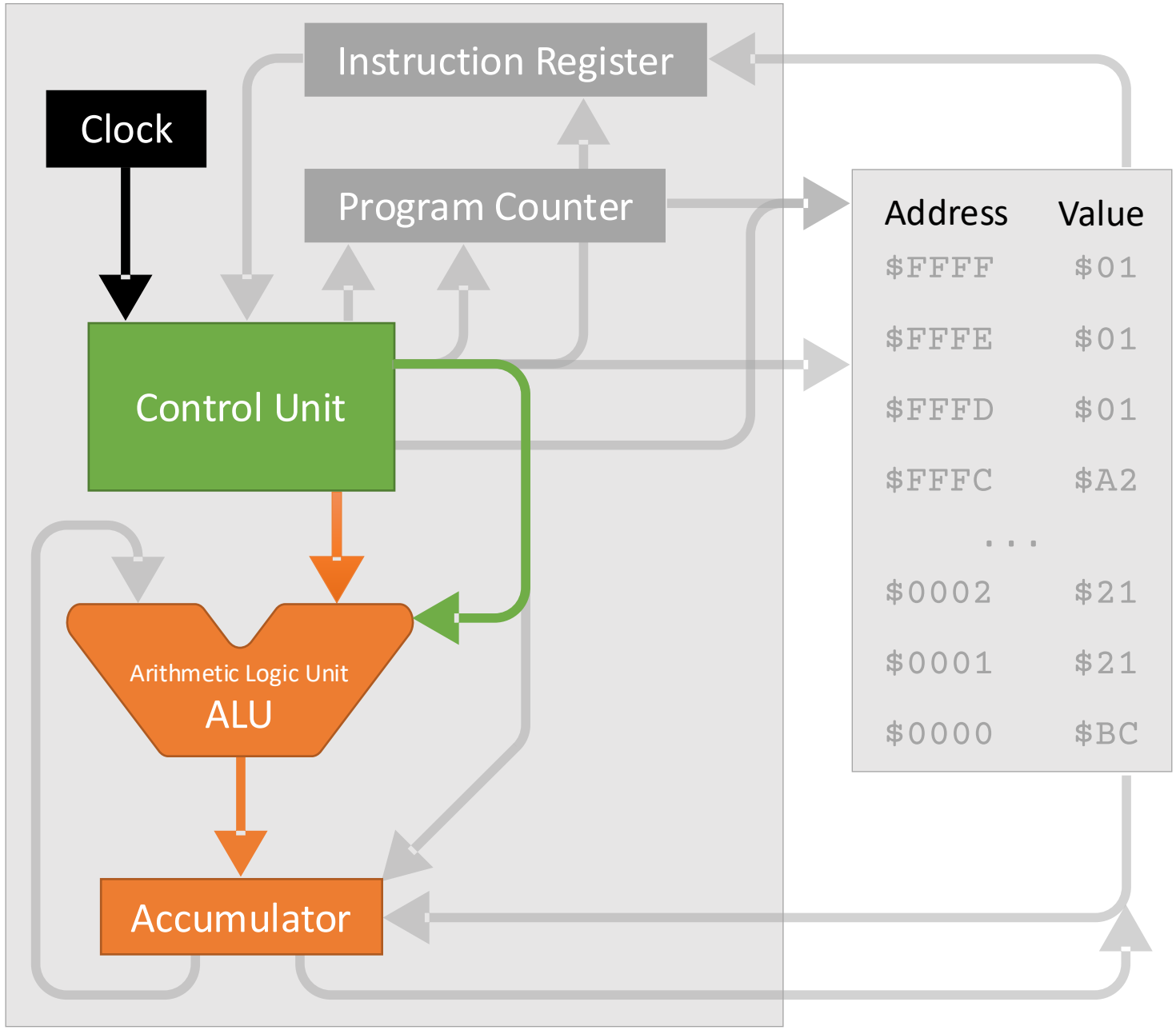
Control Bus

Address Bus

Data Bus

LDA #$05

Load accumulator with immediate value 5



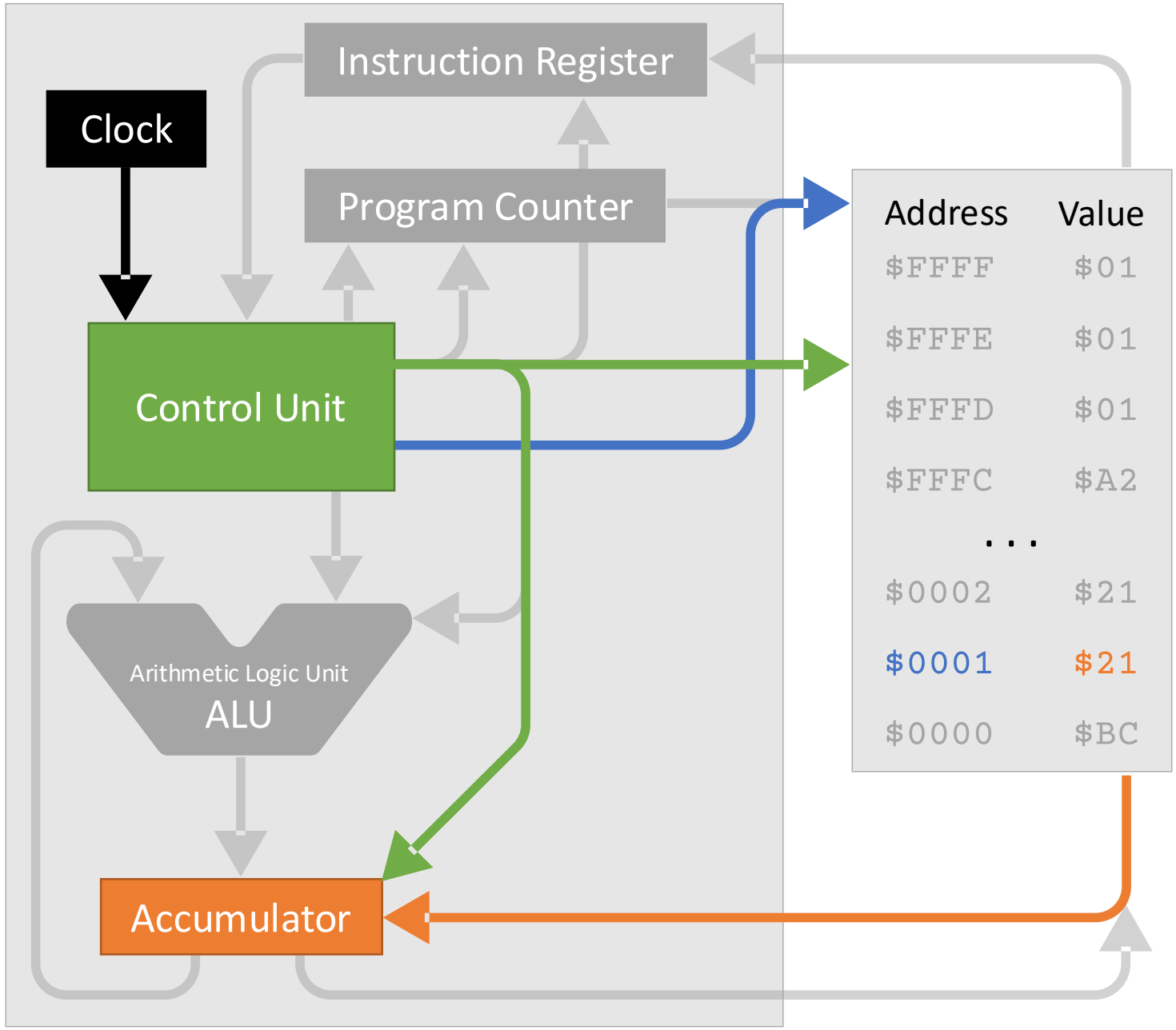
Control Bus

Address Bus

Data Bus

LDA $0001

Load accumulator value at memory address $01



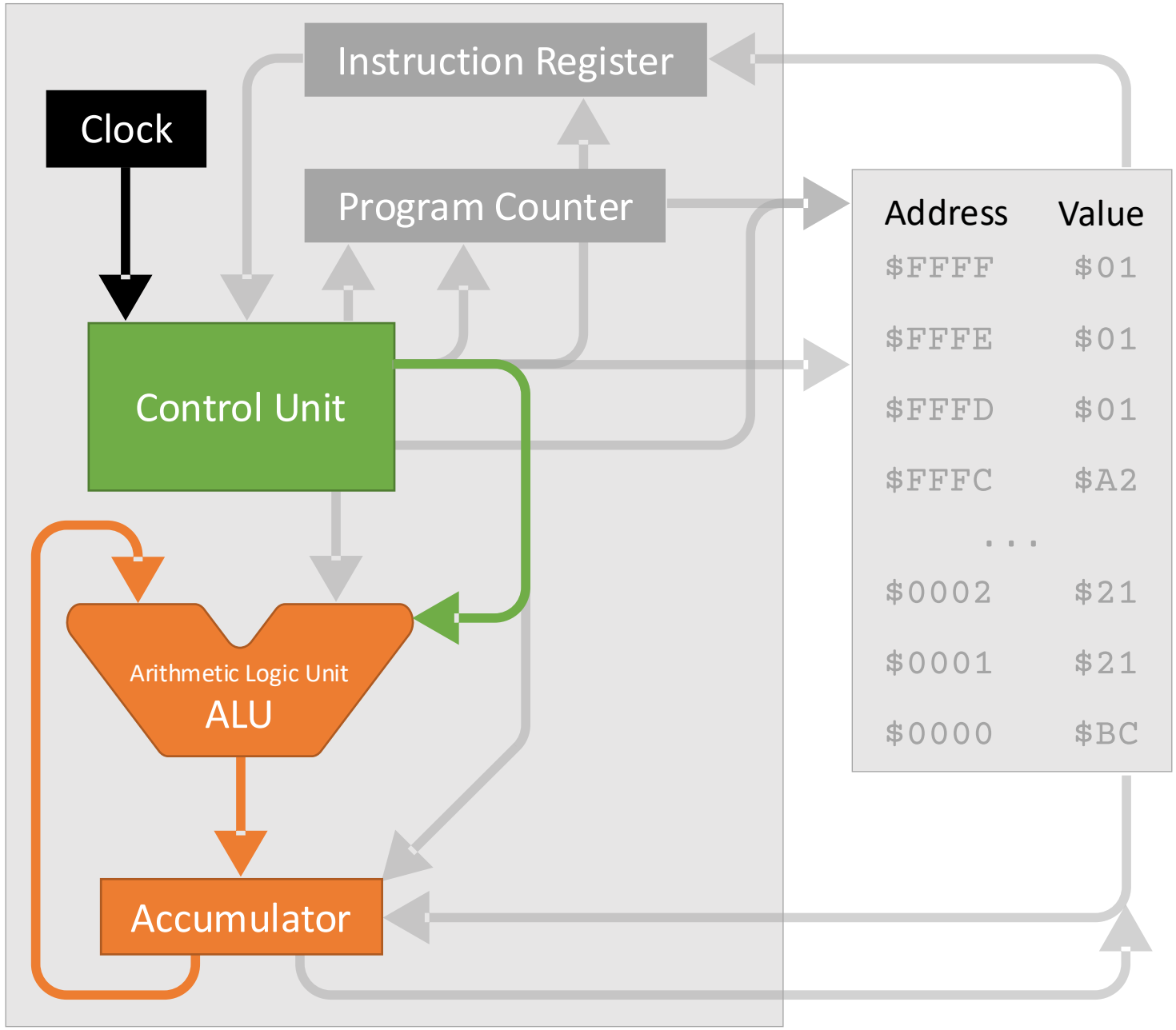
Control Bus

Address Bus

Data Bus

DECA

Decrement accumulator



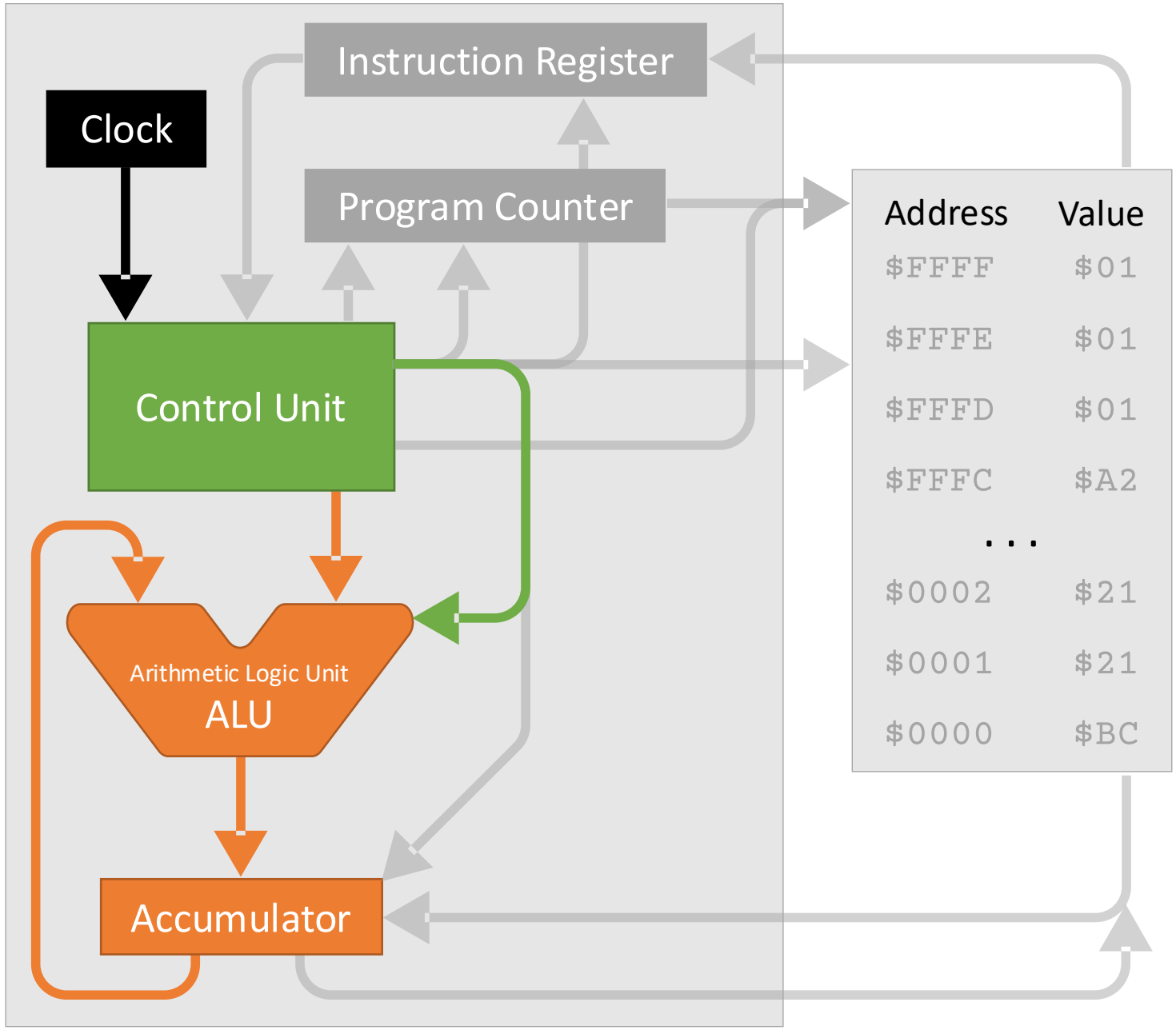
Control Bus

Address Bus

Data Bus

ADDA #$A

Add 10 to accumulator



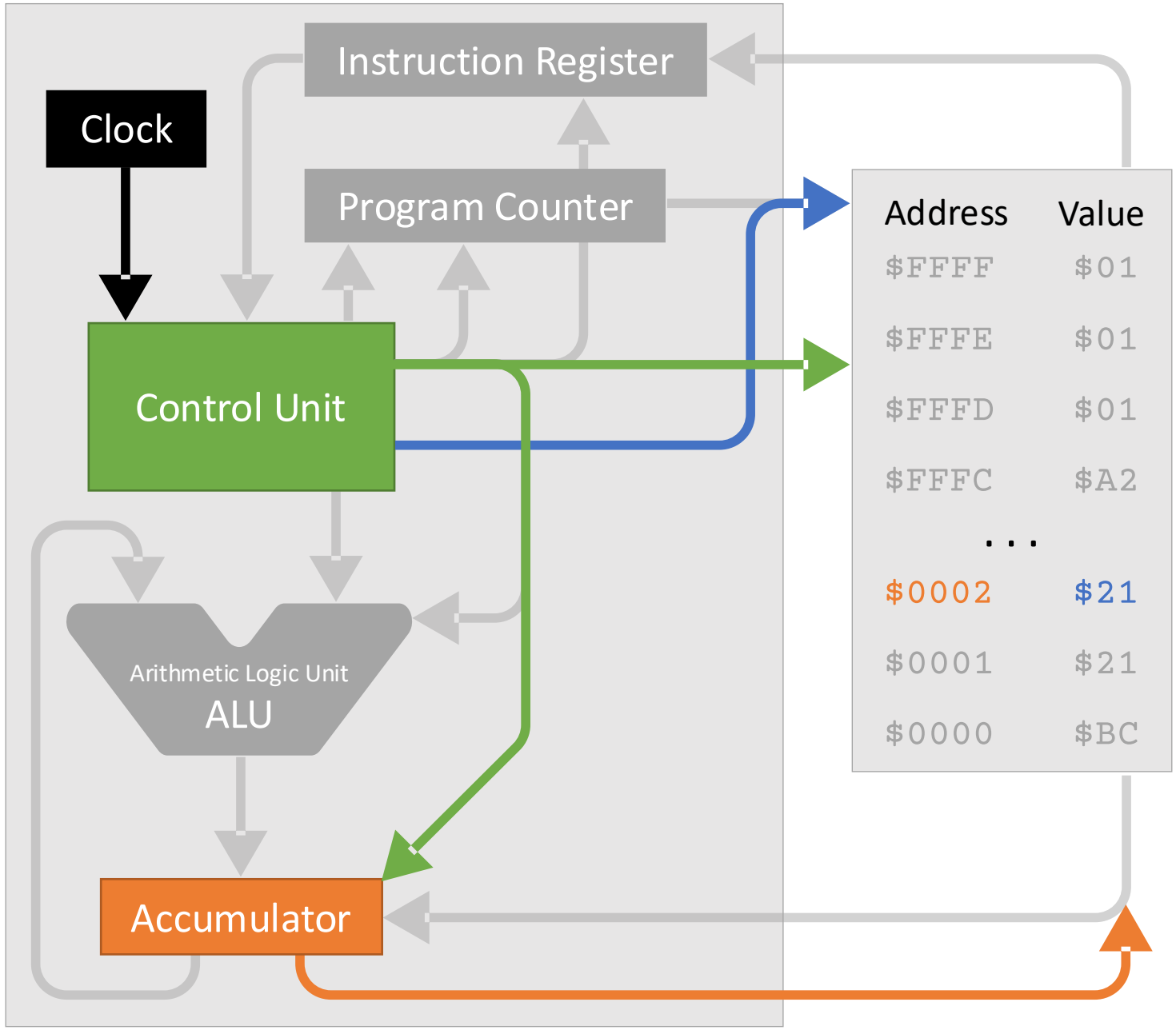
Control Bus

Address Bus

Data Bus

STA $0002

Store accumulator at memory address $02



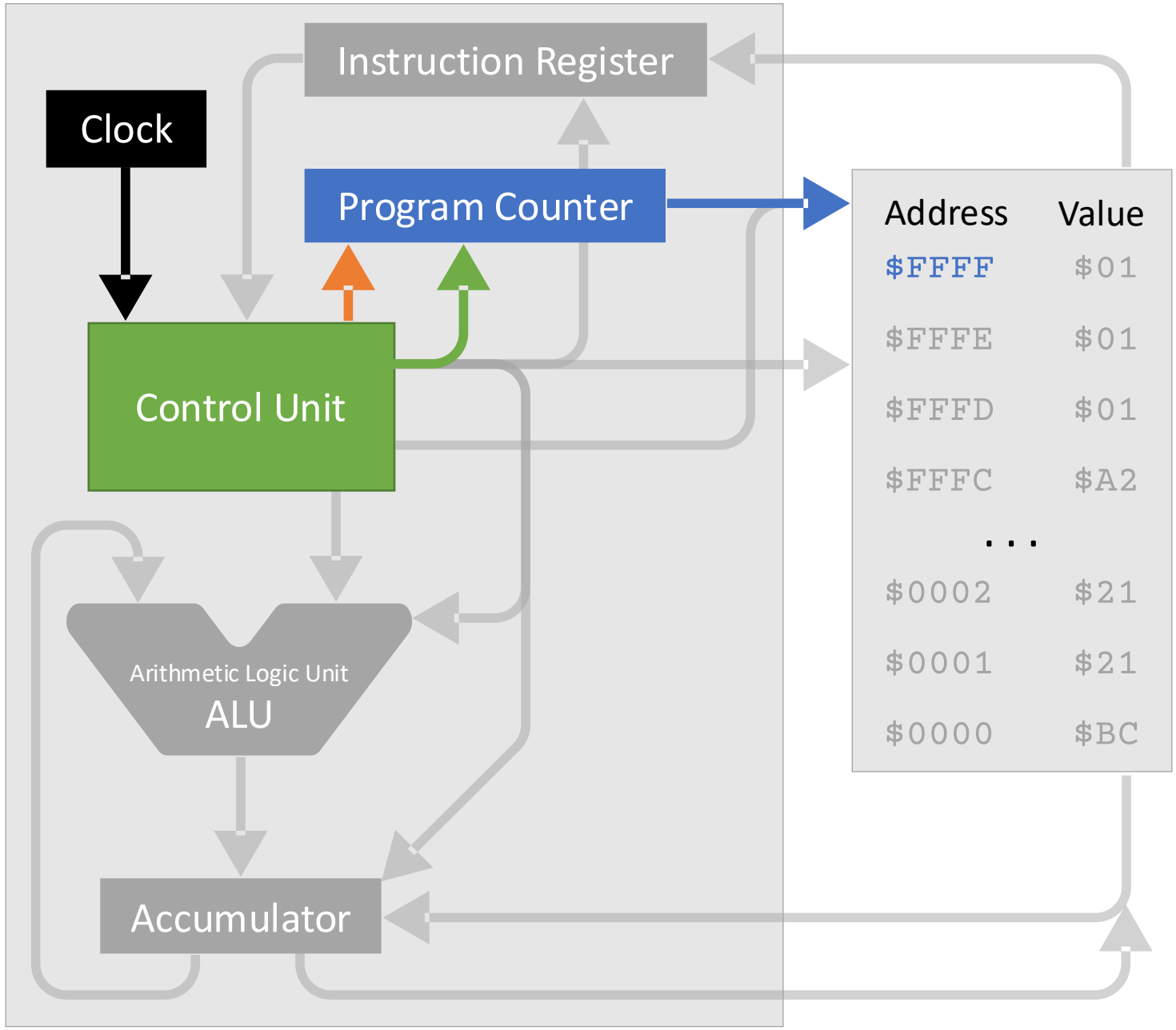
Control Bus

Address Bus

Data Bus

BRA $FFFF

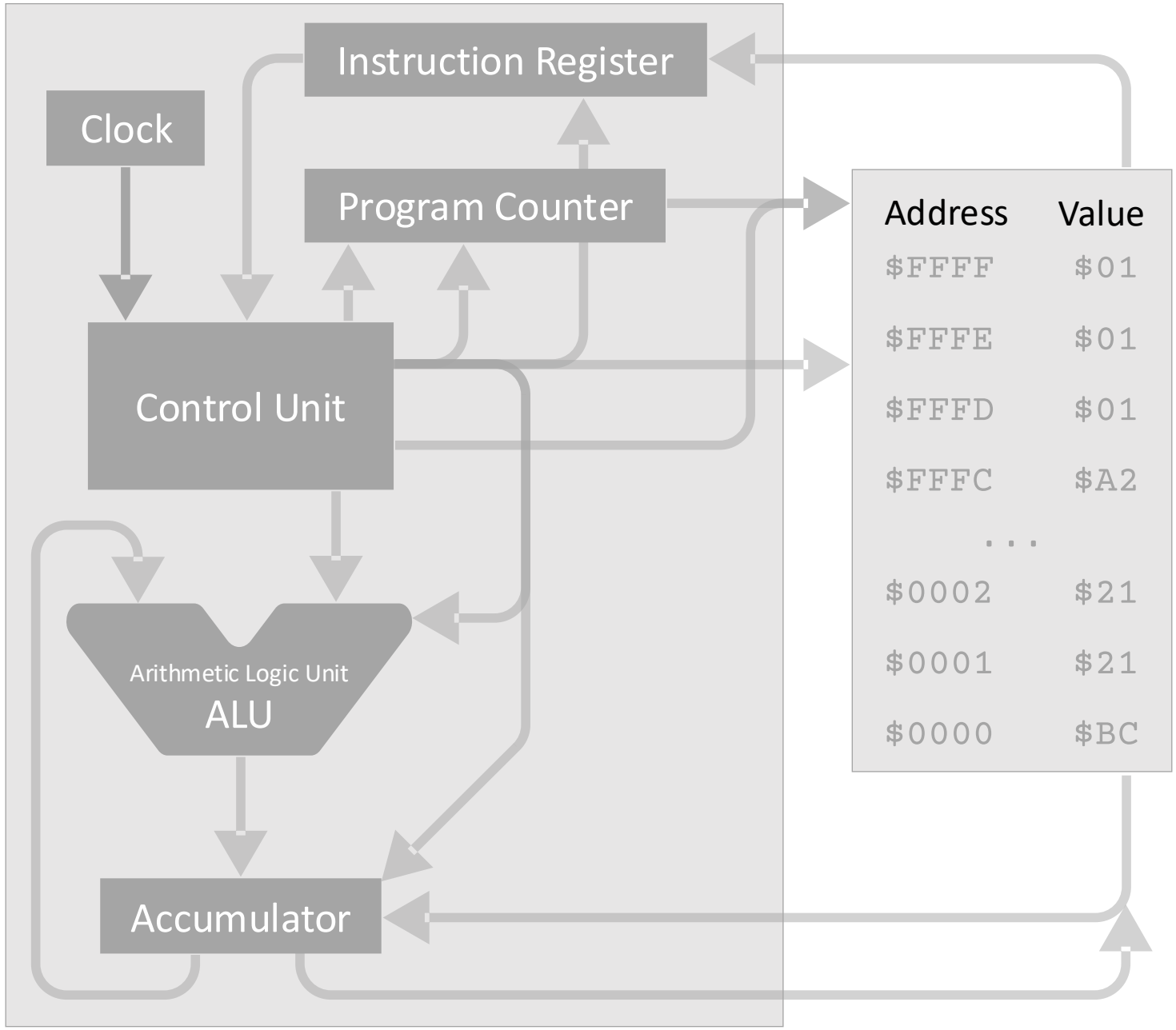
Branch to $FFFF



Control Bus

Address Bus

Data Bus

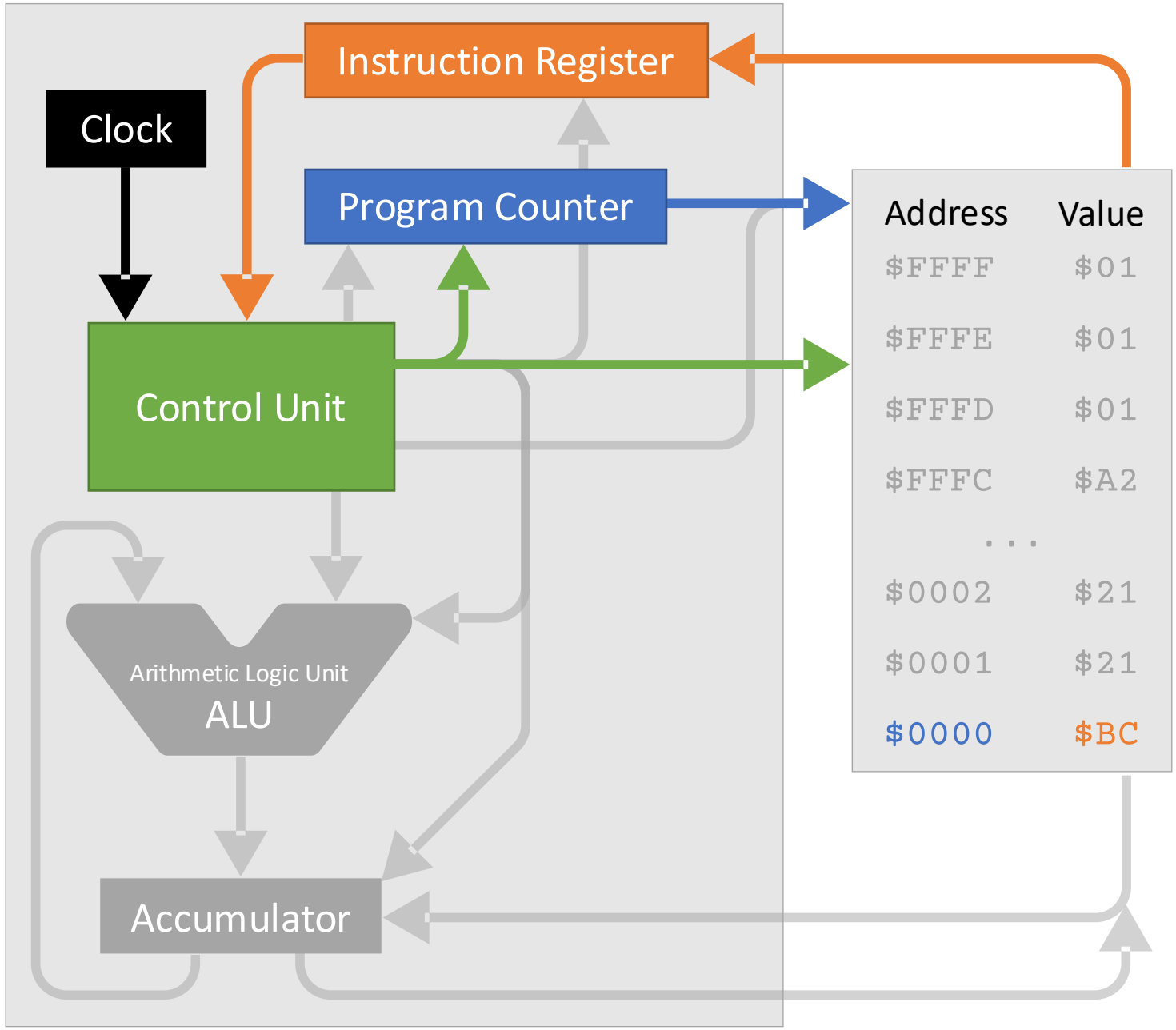


Control Bus

Address Bus

Data Bus

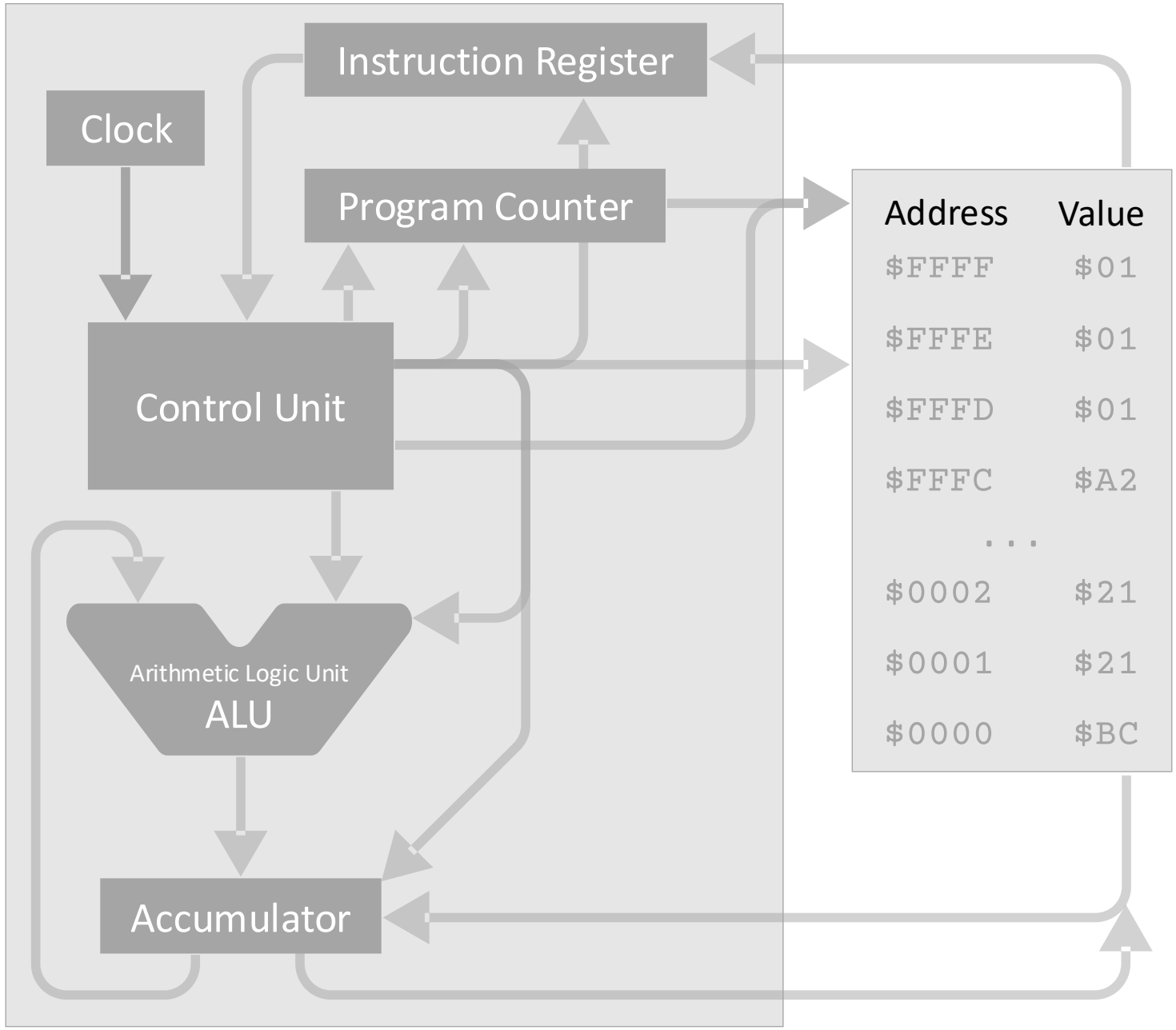
Fetch Instruction



Control Bus

Address Bus

Data Bus

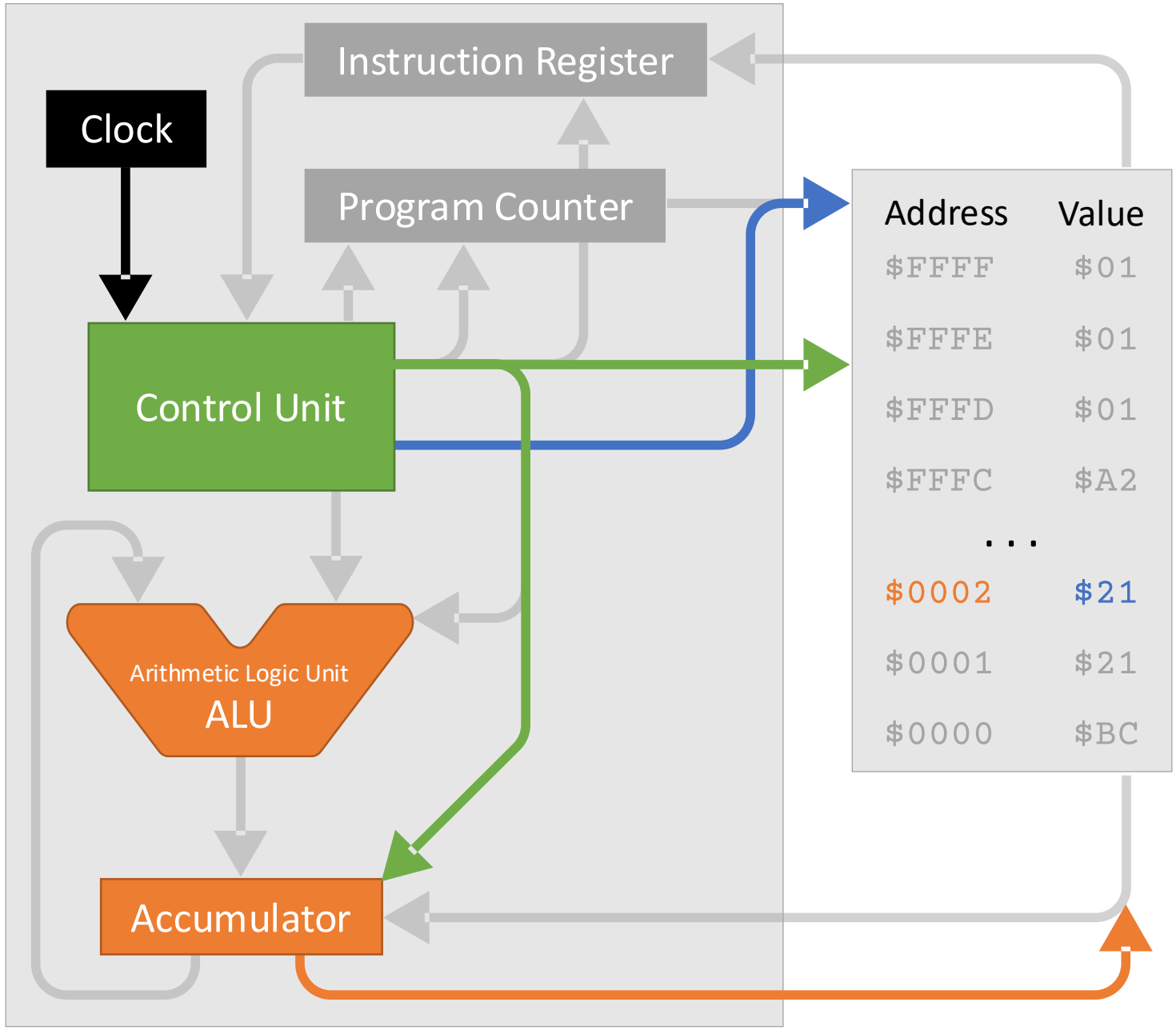


Control Bus

Address Bus

Data Bus

Execute Instruction



Control Bus

Address Bus

Data Bus